

Blockchain Hinglish Notes



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"ChatGPT, main Blockchain handwritten notes ki images bhej raha hoon. Inko dhyan se padhkar bahut hi detailed Hinglish notes banao — har topic teacher jaise explain karo, apne khud ke extra points, examples aur simple explanations bhi add karo. Diagrams ko bhi describe karo. Notes itne clear ho ki beginner bhi samajh sake. Headings, sub-headings, Hindi-English mix language

Bahut badhiya! Chalo shuru karte hain — main tumhare diye gaye Blockchain handwritten images ko dhyan se padhkar **detailed Hinglish notes** bana raha hoon.

Blockchain Basics Hinglish Notes

✓ 1 Blockchain ka Basic Concept

- **Definition:**

Blockchain ek *distributed digital ledger* hai jisme transactions ka record blocks ke form mein store hota hai. Har block ek chain se connected hota hai — isliye naam **Block + Chain = Blockchain**.

- **Key Points:**

- Sabse pehle ek transaction hota hai (jaise *Ram ne Shyam ko 1 BTC bheja*).
- Ye transaction *mempool* mein jata hai — yani pending state mein.
- Miners competition mein lagte hain transaction ko verify karne ke liye.

✓ 2 Mining & Proof of Work

- **Mining:**

Mining ka matlab hota hai naye block create karna aur usse chain mein jodna.

Miners ek *special number* (nonce) find karte hain jisse hash ka prefix kuch zero se start ho.

- **Diagram se Explanation:**

1. Transaction → prefix + prevHash + nonce → hash
2. Hash agar required pattern ko meet karta hai (e.g. starting zeros), toh block valid hota hai.
3. Bitcoin mein average 10 min lagta hai ek block mine hone mein.

- **Example:**

```
bash
```

```
prefix + nonce ==> hash  
00000bcvhjx...
```

Mining process mein random *nonce* try kiya jata hai jab tak required hash nahi milta.

✓ 3 Nonce aur Hashing

- **Nonce:**

Nonce ek random number hota hai jo miners ko guess karna hota hai.
Jaise:

```
vbnet
```

```
Transaction: Ram send 1 BTC to Shyam  
Nonce: 48206
```

Ye nonce milne ke baad hi block banega aur confirm hoga.

✓ 4 Longest Chain Rule

- Diagram mein diya hai:

```
bash
```

```
Longest valid chain hi source of truth maani jayegi.
```

Matlab agar do ya zyada chain banti hain toh sabse lambi chain ko hi valid mana jata hai.

- **51% Attack:**

Agar kisi hacker ke paas 51% computational power ho, toh woh longest chain manipulate kar sakta hai — isse network corrupt ho sakta hai.

```
yaml
```

```
e.g. 501 servers corrupted, computation power: 51%
```

Ye ek Blockchain vulnerability hai.

✓ 5 Servers & Network

- Blockchain ek distributed network hai.
- Sab nodes ya servers ek doosre se connected hain.
- Agar 499 servers honest hain aur 501 corrupted ho jayein toh corrupted servers galat chain bana sakte hain.

✓ 6 Mempool Concept

- **Mempool:**

Ye ek temporary space hai jaha saare unconfirmed transactions store hote hain.

```
yaml
```

```
mempool: 1000 transactions
```

Miners inhe pick karte hain aur naye blocks banate hain.

✓ 7 Confirmed vs Unconfirmed Transactions

- Jo block mine ho gaya aur chain mein add ho gaya woh **confirmed** hai.
- Jo abhi mempool mein pada hai woh **unconfirmed** hai.

📌 Extra Important Points

✓ Blockchain ke benefits:

- Tamper-proof ledger
- Decentralized (no single point of failure)
- Secure aur transparent transactions

✓ Blockchain ka use cases:

- Cryptocurrencies (Bitcoin, Ethereum)
- Supply chain tracking
- Smart contracts
- Voting systems

✓ Example Flow:

1. Ram sends 1 BTC to Shyam.
2. Transaction mempool mein gaya.
3. Miners compete to find nonce.
4. Valid nonce mil gaya → block ban gaya.
5. Block chain mein add ho gaya → transaction confirmed.

📌 Diagrams Explained

✓ Block Structure:

- Har block mein hota hai:
 - Previous block hash

- Current transactions
- Nonce
- Current hash

✓ Chain:

- Blocks ek chain se linked hote hain.
- Agar koi beech ka block modify ho, toh poori chain ka hash badal jata hai — isse cheating rokna easy hota hai.

✓ Summary in Hinglish

Blockchain ek secure, distributed ledger hai jisme miners *proof of work* ke through blocks banate hain. Longest valid chain ko hi sach mana jata hai. Mempool mein sabhi unconfirmed transactions store hote hain aur nonce find karna mining ka main kaam hai. Agar kisi ke paas 51% se zyada computational power ho, toh woh chain manipulate kar sakta hai — isliye network ka decentralization bahut important hai.

💡 Ye notes beginner friendly hai — agar chaho toh main isko DOC, PDF ya presentation ke form mein bhi bana dunga! Batao?